CMSC 25300/35300, STAT 27700 (Fall 2024)
Homework 1: Vectors and Matrices

Submission Instructions:

- Please submit your homework in PDF to Gradescope (which can be accessed from the course's website on Canvas);
- Please submit your code in a separate file. This can be in either a script file (.py) or a notebook (.ipynb).
- Note that you do not need to copy the problem statements in your solution, *as long as you clearly indicate the problem numbers* (e.g., 1.a, 2.c, etc). This goes for both the PDF and the submitted code.

1. Reading assignment. Read Boyd VMLS chapters 1 & 6.

2. **Matrix multiplication.** You are a UChicago student taking 3 classes this quarter, class A, class B, and class C. During first week, you spend 5 hours on class A, 7 hours on class B, and 3 hours on class C. During second week, you spend 8 hours on class A, 6 hours on class B, and 9 hours on class C. During third week, you spend 2 hours on class A, 7 hours on class B, and 12 hours on class C. During fourth week, you spend 7 hours on each class.

a) (5 points) Write this information as a matrix. What do the rows represent? What do the columns represent?

b) (3 points) Each class has a different number of students. Class A has 7 students, class B has 25 students, and class C has 6 students. Assuming all students spend the same amount of time per class per week as you do, how many total student-hours were spent on classes A, B, and C combined each week? Write a matrix-vector multiplication that calculates the total student-hours for each week.

c) (3 points) Again using matrix multiplication, how would you calculate your total workload in each class across all four weeks? (i.e. How many hours have you spent in total on class A? Class B? Class C?)

d) (3 points) Calculate the total workload (student-hours) across all weeks and classes summed for all students in classes A, B, and C (using, you guessed it, matrix multiplication).

e) (10 points) Get up and running with Python. Write a script that computes the matrix multiplications in the previous parts of this problem.

3. You are given a matrix

$$\mathbf{A} := \begin{bmatrix} 3 & 9 & 5 & 3 \\ 1 & 2 & 8 & 2 \\ 4 & 6 & 9 & 3 \\ 1 & 5 & 7 & 8 \\ 5 & 3 & 9 & 4 \end{bmatrix} \in \mathbb{R}^{5\times 4}.$$

- **a)** (3 points) Assume $\mathbf{y} = \begin{bmatrix} 10 & -5 & -1 & 2 & -8 \end{bmatrix}^\top$. Does $\mathbf{v} = \begin{bmatrix} -1 & 2 & -1 & 0 \end{bmatrix}^\top$ satisfy the equation $\mathbf{Av} = \mathbf{y}$?

- **b)** (3 points) Given a number $k \in \{1, 2, 3, 4, 5\}$, how would you construct a vector $\mathbf{v} \in \mathbb{R}^5$ such that $\mathbf{v}^\top \mathbf{A}$ is the k-th row of A?

- **c)** (3 points) How would you construct a vector $\mathbf{v} \in \mathbb{R}^5$ such that $\mathbf{v}^\top \mathbf{A}$ is 2 times the second row of $\mathbf{A}$ minus 3 times the fourth row of $\mathbf{A}$?

- **d)** (3 points) How would you construct a vector $\mathbf{v} \in \mathbb{R}^4$ such that $\mathbf{Av}$ is the last column of $\mathbf{A}$ minus the first column of $\mathbf{A}$ plus 2 times the third column of $\mathbf{A}$?

- **e)** (3 points) Given a number $k \in \{1, 2, 3, 4\}$, how would you construct a vector $\mathbf{v} \in \mathbb{R}^4$ such that $\mathbf{Av}$ is the k-th column of $\mathbf{A}$?

- **f)** (3 points) How would you construct a vector $\mathbf{v} \in \mathbb{R}^4$ such that $\mathbf{Av}$ is a times the k-th column of $\mathbf{A}$ plus b times the j-th column of $\mathbf{A}$ for some $a, b \in \mathbb{R}$ and $j, k \in \{1, \ldots, 4\}$ $(k \neq j)$?

- **g)** (8 points) In Python, write a script to verify your answers above.

4. Matrix Rank Solve the following problems:

- **a)** (5 points) This matrix has rank $= 2$:
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  $$\begin{bmatrix} 3 & -1 & 1 \\ 4 & 3 & 10 \\ -5 & 4 & 3 \\ 8 & 7 & 22 \end{bmatrix}.$$

  How can you tell? (Hint: you don't need any tools beyond what was discussed in class.)

- **b)** (5 points) What's the rank of $\mathbf{vv}^\top$, where $\mathbf{v} = \begin{bmatrix} 1 & -2 \end{bmatrix}^\top$? (Hint: you don't need any tools beyond what was discussed in class.)

5. (15 points)

You learn a model that says:

$$\hat{y}_i = w_1 x_{i,1} + w_2 x_{i,2} + w_3$$
$$\text{predicted label } = \begin{cases} +1, & \hat{y}_i > 0 \\ -1, & \text{otherwise} \end{cases}.$$

Moreover, you know that data points $(4, -2)$ and $(-1, 1)$ are both on the decision boundary, whereas the predicted label for $(4, 0)$ is $+1$. Please find all possible weight vectors $\mathbf{w}$ that satisfy the above conditions. Do different $\mathbf{w}$ correspond to different decision boundaries? Draw a plot and identify all the data points $\mathbf{x}$, for which your model will predict the label as -1.

6. Polynomials using linear models.

Suppose we observe pairs of scalar points (z_i, y_i), $i = 1, \ldots, n$. Imagine these points are measurements from a scientific experiment. The variables z_i are the experimental conditions and the y_i correspond to the measured response in each condition. Suppose we wish to fit a degree $d < n$ polynomial to these data. In other words, we want to find the coefficients of a degree d polynomial p so that $p(z_i) \approx y_i$ for $i = 1, 2, \ldots, n$. We want to use a linear model.

a) (5 points) Suppose p is a degree d polynomial. Write the general expression for $p(z) = y$.

b) (5 points) Express the $i = 1, \ldots, n$ equations as a system in matrix form $\boldsymbol{X}\boldsymbol{w} = \boldsymbol{y}$. Specifically, what is the form/structure of $\boldsymbol{X}$ in terms of the given z_i.

c) (15 points) Write a Python script to generate a plot of a polynomial given $\mathbf{w}$ and points $z_1, \ldots, z_n$. Here is some starter code and you are asked to fill in the TODOs:

```
import numpy as np
import scipy.io as sio
import matplotlib.pyplot as plt

# n = number of points
# z = points where polynomial is evaluated
# p = array to store the values of the interpolated polynomials
n = 100
z = np.linspace(-1, 1, n)

d = 4 # degree
w = np.random.rand(d)
X = np.zeros((n,d))

# TODO: generate X-matrix

# TODO: evaluate polynomial at all points z,
# and store the result in p
```

```
# do NOT use a loop for this

# plot the datapoints and the best-fit polynomials
plt.plot(z, p, linewidth=2)
plt.xlabel('z')
plt.ylabel('y')
plt.title('polynomial with coefficients w = %s' %w )
plt.show()
```